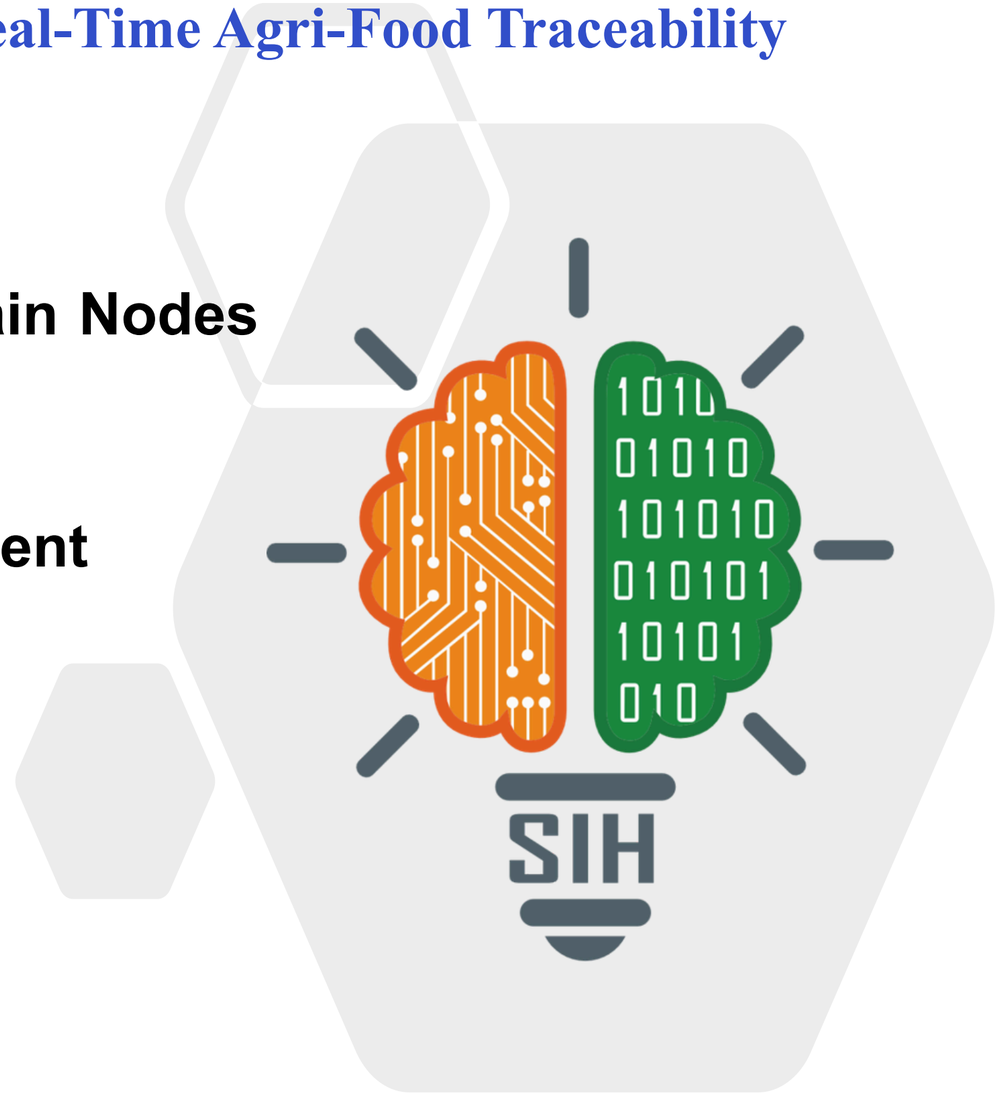
# SMART INDIA HACKATHON 2026

## Low-cost IoT-Blockchain System for Secure and Real-Time Agri-Food Traceability

- Problem Statement ID – **SIH26232**
- Problem Statement Title - **Low-Cost IoT Block chain Nodes for Farm-to-Fork Traceability**
- Theme**- Agriculture, FoodTech & Rural Development**
- PS Category**- Hardware**
- Team ID- **177933**
- Team Name (Registered on portal) **- Ekagra**



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# PROPOSED SOLUTION

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## The Problem



Agri-food supply chains lack an affordable, reliable, and tamper-resistant way to continuously monitor and trace food from farm to market, especially in remote areas and during cold-chain disruptions.

### Major Gaps

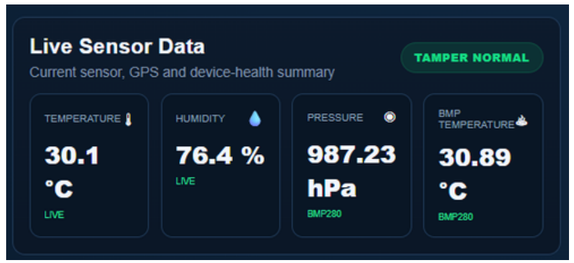
- Poor End-to-End Traceability
- Food Fraud & Data Manipulation
- Unreliable Rural Connectivity
- Lack of Tamper-Evident Records
- Limited Monitoring of Product Conditions
- Fragmented Supply-Chain Data
- High Cost of Existing Solutions
- Cold-Chain Monitoring Gaps

## Our Solution

A low-cost IoT node that travels with agricultural produce, continuously records its environmental condition, motion and location, stores every record locally on an SD card even without internet, and later syncs it to a backend where records are hash-chained and anchored on a blockchain, so no one can secretly alter the transit history.

- Temperature
- Humidity
- GPS Location
- Motion & Vibration
- Tamper Detection
- Ethylene Level
- Atmospheric Pressure
- Timestamp & ID

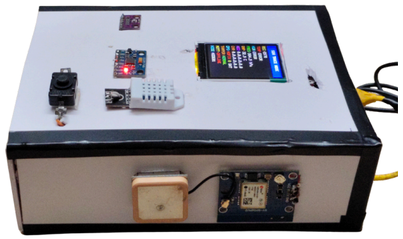
### What the node monitors??




Screenshot from Dashboard

## Our Prototype





## Innovation & Uniqueness

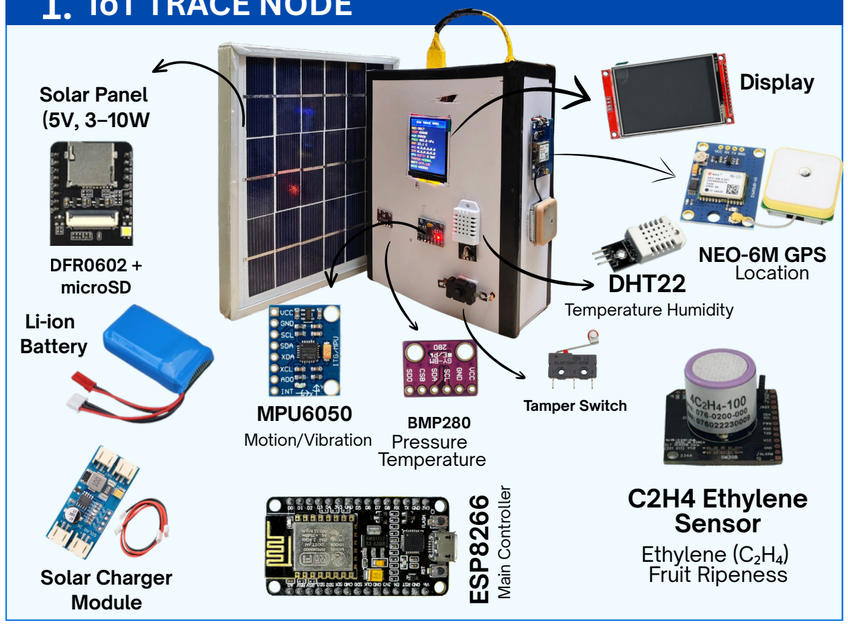
- **Dual-Controller Architecture –** Separate sensing and storage for reliable data logging.
- **Offline Store-and-Forward –** Records are saved locally and synced when connectivity returns.
- **Multi-Level Tamper Evidence –** Physical switch + SHA-256 hash chaining + blockchain anchoring.
- **Complete Condition Monitoring –** Temperature, humidity, pressure, vibration, GPS, time and ethylene.
- **Low-Cost Blockchain Integration –** Only the batch Merkle root is stored on-chain.
- **Modular & Affordable –** Sensors can be added as required, keeping the system suitable for SMEs.
- **QR-Based Verification –** Buyers and regulators can independently verify consignment history.

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# TECHNICAL APPROACH

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## 1. IoT TRACE NODE




## 2. DATA ACQUISITION & LOCAL STORAGE

**Sesor Reading As Trace Node**

**Node ID**: NODE_01
**Record ID**: 2207
**Timestamp**: 2026-09-30 23:32:48
**Uptime**: 460339 s
**Temperature**: 27.1 °C
**Humidity**: 81.3 %
**Pressure**: 985.3 hPa
**Ethylene**: 0.48 ppm
**BMP Temp**: 27.0 °C
**GPS**: 24.055014, 84.174537
**Motion**: ACC, GYRO values
**Tamper**: NORMAL
**Battery**: 3.8 V (Charging)
**Solar**: 5.1 V / 120 mA

**ONLINE**

4G
Wi-Fi / 4G Available

MQTT
Publish to Broker

**OFFLINE**

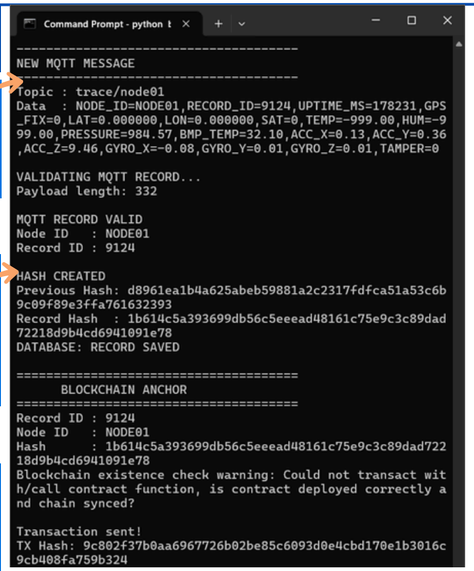
SD
Data Encryption
AES-256 Encryption
Encrypted Backup Storage on SD

## 3. BACKEND ARCHITECTURE

MQTT **Receive Records**

**Python Backend**
(API + Processing)
Validate, parse, store, hash, synchronize, and map each record

**SQLite Database**
Stores complete sensor, GPS, ethylene, motion, tamper, timestamp, and node data




**Screenshot**: Live Python backend processing of an MQTT trace record, including validation, hash generation, database storage

## 4. BLOCKCHAIN & DATA INTEGRITY

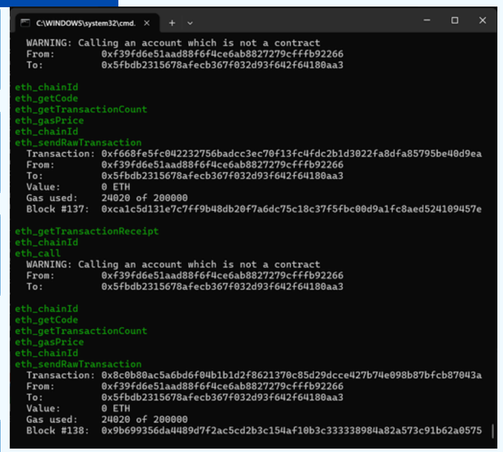
**Canonical Record**
Selected fields

**Previous Hash + Current Record**

**SHA-256 Hash**
Generates a unique cryptographic hash

**Record Hash**
Unique fingerprint of the record

**Blockchain Anchor**
Stores hash and metadata as immutable proof




**Screenshot: Blockchain transaction generated for anchoring the traceability record and providing tamper evident data integrity.**

**HASH CHAIN**

**Record 2021**
Initial Record
↓ SAH-256
HASH 1

**Record 2022**
Hash 1 + Current record
↓ SAH-256
HASH 2

**Record 2023**
Hash 2 + Current record
↓ SAH-256
HASH 3

**DEVICE SECURITY**

**Hardware-backed Identity**
Secure Element / TPM Stores device private key

**Device Authentication**
Authenticates the IoT node to the blockchain network

**Secure Communication**

## 5. DATA VERIFICATION PROCESS

**Retrieve Stored Record**
From SQLite/SD (OFF-CHAIN)

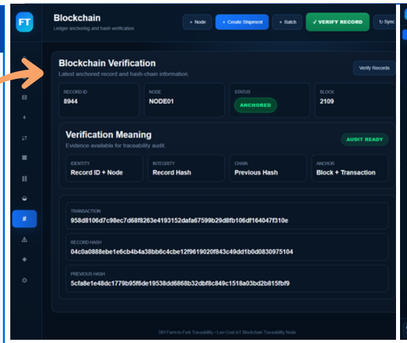
**Recalculate SHA-256**
Using previous hash + current record

**Compare with Blockchain**
Match with on-chain hash

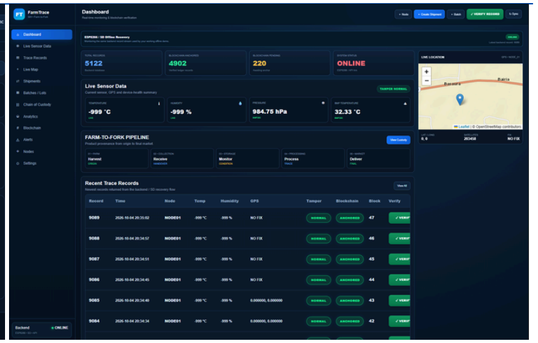
**VERIFIED**
Record is authentic and not modified

**MODIFIED**
Hash mismatch (record is tampered)

**Prototype Explaination Video Link**




**Screenshot : Blockchain Verification for particular record**




**Monitoring Dashboard Homepage**

## TRACEABILITY & TRANSPARENCY

**Monitoring Dashboard**
Authorized Stakeholders

**QR Code Access**
View product journey and verification status

**Multi-Participant Access**
Farm, Processor, Transporter, Retailer

**Audit & Compliance**
Regulators / Certifiers

# FEASIBILITY AND VIABILITY



**₹ 2K-3K**
**Prototype Cost (Local Parts)**

**Secure and Immutable**
(using Blockchain latency>100ms)

## Technical Feasibility

- **Proven Prototype –** Core sensors, GPS, UART, SD storage, and trace logging are already tested.
- **Off-the-Shelf Hardware –** Uses affordable, well-supported commercial modules and open-source libraries.
- **Reliable Low-**Bandwidth Communication – MQTT suits unreliable rural networks.
- **Low-Cost Cryptography –** SHA-256 hashing can run directly on the ESP8266/ESP32.
- **Scalable Backend –** MQTT, databases, APIs, and blockchain can scale to multiple nodes.
- **Easy Upgrade Path –** Ethylene sensing, 4G, solar power, and rugged enclosures can be added modularly.

## Economical Feasibility

- **Low Prototype Cost –** Current prototype costs only ₹2,00–3,300.
- **Lower Cost at Scale –** Custom PCB and bulk components can further reduce unit cost.
- **Reusable Hardware –** Nodes can be reused across multiple shipments.
- **Flexible Business Model –** Rental per shipment or device purchase with subscription options.
- **Low Operating Cost –** Low-data MQTT, solar charging, and minimal blockchain transactions reduce running costs.
- **Strong ROI Potential –** Preventing a single spoiled or rejected shipment can offset the device cost.

## Feasibility

## Maintenance Feasibility

- Modular sensor architecture allows individual sensors to be replaced without replacing the complete node.
- Firmware can be updated remotely when connectivity is available.
- Battery and sensor health can be monitored through the dashboard.
- Solar charging reduces the frequency of manual battery replacement.
- Local data buffering prevents data loss during temporary maintenance or network failures.
- Rugged enclosure protects electronics from dust, moisture and vibration.

## Operational Feasibility

- **Plug-and-play node** can be attached to crates, containers or transport vehicles.
- Automatic sensing and logging minimizes operator intervention.
- Data continues to be recorded during network/cellular dropouts.
- Automatic synchronization occurs when connectivity is restored.
- Dashboard provides real-time visibility of: Temperature Humidity EthyleneLocation Tamper events Shipment history
- Threshold-based alerts can identify environmental excursions during transit.

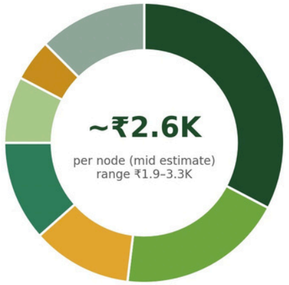
## Viability

## Environmental Viability

- Solar energy harvesting reduces dependence on disposable/rechargeable battery replacement.
- Low-power operation minimizes energy consumption. Reusable hardware reduces the need for disposable data loggers.
- Local processing reduces unnecessary continuous data transmission.
- Rugged, repairable sensor modules can extend device lifetime and reduce electronic waste.
- Supports better food-chain monitoring, potentially helping identify temperature excursions and spoilage conditions earlier, reducing avoidable food losses.

## Social Viability

- Low-voltage electronics reduce electrical hazards during operation.
- Enclosed electronics protect sensors and circuitry from moisture and accidental contact.
- Tamper detection records unauthorized opening or physical interference.
- Encrypted storage protects sensitive supply-chain information.
- Cryptographic integrity checks make unauthorized modification of historical records detectable.
- The node performs monitoring and traceability, without directly controlling hazardous industrial equipment




- DFR0602 ESP32 storage bridge ₹850 (33%)
- NEO-6M GPS ₹500 (19%)
- ESP8266 NodeMCU ₹300 (12%)
- DHT22 ₹300 (12%)
- MPU6050 ₹200 (8%)
- BMP280 ₹125 (5%)
- microSD, switch & wiring ₹325 (12%)

## How we are different from existing Market Devices

| Features | Available USB Loggers | Premium Real Time Tracker | Our Solution |
|---|---|---|---|
| **Cost** | Low | High, plus recurring subscription | **Low, rent-per-trip option** |
| **Location tracking** | No | Yes | **Yes (GPS)** |
| **Works without network** | Yes, but data read only at destination | Limited buffer | **Full SD storage + auto sync** |
| **Tamper-proof data** | Data can be edited after export | Trust depends on vendor's cloud | **Hash chain + blockchain, independently verifiable** |
| **Physical tamper detection** | No | Some models | **Yes** |
| **Ethylene monitoring** | No | Rare | **Yes (final build)** |
| **Reusable** | Some | Often single-use or leased | **Reusable** |
| **Customizable / open** | No | No, closed ecosystem | **Modular, crop profiles** |

# IMPACT AND BENEFITS



## Impacts

- End-to-end digital traceability across the farm-to-fork supply chain.
- Blockchain-based data integrity through tamper-evident records and **cryptographic hashes.**
- Real-time supply-chain visibility using **IoT** sensor and **GPS telemetry.**
- Post-harvest loss reduction through continuous environmental-condition monitoring.
- Improved food-safety monitoring through temperature, humidity and gas-condition records.
- Supply-chain accountability through timestamped custody and location events.
- Increased consumer transparency through QR-based product verification.

## Benefits

- **Low-cost** IoT deployment using an ESP32-based node.
- **Offline-first operation** with local microSD data storage.
- Multi-network connectivity using Wi-Fi.
- Solar-powered operation for field and remote deployments.
- Ruggedized deployment using an **IP65**-type enclosure.
- Scalable architecture supporting multiple IoT nodes.
- **QR-enabled access** to verified product-journey information.
- Permissioned blockchain enables controlled access to supply-chain records.
- Modular architecture allows sensors and communication modules to be adapted to different supply-chain stages.

## Key Outcomes



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# RESEARCH AND REFERENCES

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## Research

1. **Temperature and humidity excursions**: Following the Q10 principle, a produce's respiration rate roughly doubles to triples for every 10 °C rise in temperature, sharply reducing shelf life. High relative humidity with temperature fluctuation causes condensation, which promotes fungal and bacterial growth, while low humidity causes moisture loss, wilting and weight loss. Short, undetected excursions during loading, unloading or reefer failure often go unrecorded in conventional logging.
2. **Mechanical damage from shock and vibration**: Poor road conditions, overloading and rough handling expose produce to impact, compression and vibration. These cause bruising, cracking and internal tissue damage that may not be visible at delivery but trigger rapid decay later. Accelerometer-based monitoring (g-force spikes, vibration frequency) can identify when and where damage occurred along the route.
3. **Ethylene accumulation in closed transport**: Climacteric fruits such as banana, mango and tomato release ethylene, a ripening hormone that accumulates in closed containers. Elevated ethylene causes premature ripening, softening and yellowing, and mixed loads spread it to ethylene-sensitive produce. Monitoring ethylene concentration enables timely ventilation, segregation or use of ethylene scrubbers.
4. **IoT-based cold-chain monitoring limitations**: IoT sensor nodes using protocols such as MQTT, LoRa and NB-IoT/LTE-M enable real-time condition monitoring with low bandwidth and power. However, rural transit routes have frequent cellular dropouts, causing data loss or gaps. Research shows the need for edge storage with store-and-forward synchronization, energy-efficient duty cycling and reliable timestamping using GNSS or RTC during connectivity loss.
5. **Blockchain for agri-food traceability**: Blockchain provides a decentralized, immutable, append-only ledger where each block is cryptographically linked using hash functions such as SHA-256, making historical records tamper-evident. Frameworks like AgriBlockIoT and Hyperledger Fabric-based supply chains show improved transparency and trust among farmers, transporters, retailers and regulators. Storing only Merkle roots on-chain, while keeping raw data off-chain, improves scalability and reduces transaction cost and latency.
6. **The oracle problem in blockchain-IoT systems**: Blockchain guarantees that data cannot be changed after it is written, but it cannot verify whether the data entering it was genuine. This is known as the oracle problem. If sensor data is manipulated at the device, during offline storage or in transmission before reaching the ledger, blockchain only preserves falsified records. Research highlights the need for device-level security such as hash chaining at the edge, sequential record IDs, device authentication (HMAC/ECDSA).

| Features | Basic USB Loggers | Real time Tracker | Our Solution |
|---|---|---|---|
| Low Cost & affordable for SME's | ✓ | ✗ | ✓ |
| GPS Location Tracking | ✗ | ✓ | ✓ |
| Offline storage with Auto-sync | ✗ | ✗ | ✓ |
| Tamper-Evident Records | ✗ | ✗ | ✓ |
| Physical Tamper Detection | ✗ | ✓ | ✓ |
| Shock and Vibration Monitoring | ✗ | ✓ | ✓ |
| Ethylene Gas Monitoring | ✗ | ✗ | ✓ |
| Reusable across many trips | ✓ | ✗ | ✓ |
| **Total Features Covered** | 2 / 8 | 3 / 8 | 8 / 8 |

## Key References


- **Kaur, A. et al. (2022)**. Adaptation of IoT with Blockchain in Food Supply Chain Management: An Analysis-Based Review. Sensors, MDPI. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9655358/
- IBM Blockchain Collaboration for Food Traceability — The Manufacturer. https://www.themanufacturer.com/articles/ibmsblockchain-collaboration-strengthens-food-traceability/
- Blockchain-Assisted IoT Framework for Reliable Monitoring and Predictive Analytics in Perishable Food Supply Chains
  **Aditya Gupta, Vibha Jain (06 January 2026)**
  **https://doi.org/10.1111/jfpe.70333Digital Object Identifier (DOI)**


## Useful Links

## Feature Coverage v/s Market Devices

